

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8				Indicative content Method each halogen will be added to each halide solution in order to see if a reaction occurs Observations • colour change in sodium iodide with both chlorine and bromine • colour change in sodium bromide with chlorine • no reaction when iodine is added to sodium chloride or sodium bromide or when bromine is added to sodium chloride Conclusions • chlorine displaces both bromine and iodine from bromide/iodide solutions • chlorine is therefore most reactive • bromine displaces iodine from iodide solution and is therefore more reactive than iodine • iodine does not displace chlorine or bromine and is therefore least reactive • more reactive halogens displace less reactive halogens from solution • trend in reactivity - chlorine > bromine > iodine Equations • $\text{Cl}_2 + 2\text{NaBr} \rightarrow 2\text{NaCl} + \text{Br}_2$ • $\text{Cl}_2 + 2\text{NaI} \rightarrow 2\text{NaCl} + \text{I}_2$ • $\text{Br}_2 + 2\text{NaI} \rightarrow 2\text{NaBr} + \text{I}_2$	6			6		4